

```

#include <iostream>
#include <iomanip>
#include <string>
using namespace std;

const int ROW = 5;
const int COL = 10;

struct Seat {
    bool booked;
    string passenger;
};

Seat seats[ROW][COL];

void displaySeats() {

    cout<<"\nTRAIN SEAT MAP\n\n";

    cout<<" ";
    for(int i=1;i<=COL;i++){

        cout<<setw(3)<<i;
    }
    cout<<endl;

    for(int i=0;i<ROW;i++){

        cout<<char('A'+i)<<" ";

        for(int j=0;j<COL;j++){

            if(seats[i][j].booked)
                cout<<setw(3)<<"X";

            else
                cout<<setw(3)<<"- ";

        }

        cout<<endl;
    }
}

void bookSeat(){

    char row;
    int col;

    cout<<"Enter Row (A-E): ";
    cin>>row;

    cout<<"Enter Column (1-10): ";
    cin>>col;

    if(row<'A' || row>'E' || col<1 || col>10){
        cout<<"Invalid seat selection\n";
        return;
    }

    int r = row - 'A';

```

```

int c = col - 1;

if(seats[r][c].booked){
    cout<<"Seat already booked\n";
    return;
}

string name;
cout<<"Enter Passenger Name: ";
cin>>name;

seats[r][c].booked = true;
seats[r][c].passenger = name;

cout<<"\nTicket Confirmed!\n";
cout<<"Passenger: "<<name<<endl;
cout<<"Seat: "<<row<<col<<endl;
}

void cancelSeat(){

    char row;
    int col;

    cout<<"Enter Row to cancel (A-E): ";
    cin>>row;

    cout<<"Enter Column to cancel (1-10): ";
    cin>>col;

    int r = row - 'A';
    int c = col - 1;

    if(!seats[r][c].booked){
        cout<<"Seat is already empty\n";
        return;
    }

    seats[r][c].booked = false;
    seats[r][c].passenger = "";

    cout<<"Seat cancelled successfully\n";
}

int main(){

    int choice;

    while(true){

        cout<<"\n===== TRAIN TICKET BOOKING SYSTEM =====\n";
        cout<<"1. Display Seats\n";
        cout<<"2. Book Seat\n";
        cout<<"3. Cancel Seat\n";
        cout<<"4. Exit\n";

        cout<<"Enter choice: ";
        cin>>choice;

        switch(choice){

```

```
case 1:
    displaySeats();
    break;

case 2:
    bookSeat();
    break;

case 3:
    cancelSeat();
    break;

case 4:
    cout<<"Thank you!\n";
    return 0;

default:
    cout<<"Invalid option\n";
}

}

}
```